

ADDITIONAL PROBLEM SHEET 5: SOLUTIONS

1. Acetabularia

(a) The equations reduce to

$$\begin{aligned}\frac{\partial u_1}{\partial t} &= \nabla^2 u_1 + F_u u_1 + F_v v_1, \\ \frac{\partial v_1}{\partial t} &= D \nabla^2 v_1 + G_u u_1 + G_v v_1.\end{aligned}$$

(b) This eigen-equation can be written as

$$\frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2} + k^2 \psi = 0. \quad (1)$$

The periodicity of the θ coordinate implies we can write the solution ψ as a Fourier series,

$$\psi(r, \theta) = \sum_{n=1}^{\infty} a_n(r) \cos(n\theta) + b_n(r) \sin(n\theta), \quad (2)$$

so that now we require the a and b to satisfy

$$\frac{d^2 a_n}{dr^2} + \frac{1}{r} \frac{da_n}{dr} + \left(k^2 - \frac{n^2}{r^2}\right) a_n = 0, \quad (3)$$

and the same for b_n . This is Bessel's equation and it has two linearly independent solutions,

$$J_n(kr) \text{ and } Y_n(kr), \quad (4)$$

the Bessel functions of the first and second kind (you will not be required to know what they look like). Now we have solutions in the form

$$[A_n J_n(kr) + B_n Y_n(kr)] \cos(n\theta) + [C_n J_n(kr) + D_n Y_n(kr)] \sin(n\theta), \quad (5)$$

with A_n, B_n, C_n and D_n constants.

Each part is linearly independent so they have to individually satisfy the boundary conditions. First at $r = R_i$ we apply the no flux condition,

$$A J'_n(kR_i) + B Y'_n(kR_i) = 0 \implies B = -A J'_n(kR_i) / Y'_n(kR_i), \quad (6)$$

so

$$a_n = A \left(J_n(kr) - Y_n(kr) \frac{J'_n(kR_i)}{Y'_n(kR_i)} \right). \quad (7)$$

The outer boundary condition is $a_n(R_o) = 0$ (since we need $u_0 + \varepsilon u_1 = u_0$ on this boundary, again similar for v). Thus

$$J_n(kR_o) Y'_n(kR_i) - Y_n(kR_o) J'_n(kR_i) = 0. \quad (8)$$

It is this equation which determines the permissible values for k . Thus ψ takes the form

$$\psi(r, \theta) = \sum_{n=0}^{\infty} \sum_i \{ [A_{ni}J_n(k_{ni}r) + B_{ni}Y_n(k_{ni}r)] \cos(n\theta) + [C_{ni}J_n(k_{ni}r) + D_{ni}Y_n(k_{ni}r)] \sin(n\theta) \}, \quad (9)$$

where the k_{ni} are the solutions to eq. (8) for a given n and $A_{ni}, B_{ni}, C_{ni}, D_{ni}$ are constants.

(We haven't shown it but there are a countable infinity of solutions for each n . Here, the second sum just represents all such solutions for a given n .)

(c) We expand

$$a_n(r) = c_0 + c_1\varepsilon(r - R_i) + \mathcal{O}(\varepsilon^2). \quad (10)$$

We also expand $1/r^2$ and $1/r$ about R_i to get

$$\frac{1}{r^2} = \frac{1}{R_i^2} - \frac{2\varepsilon}{R_i^3}(r - R_i) + \mathcal{O}(\varepsilon^2), \quad (11)$$

$$\frac{1}{r} = \frac{1}{R_i} - \frac{\varepsilon}{R_i^2}(r - R_i) + \mathcal{O}(\varepsilon^2). \quad (12)$$

At $\mathcal{O}(1)$ we have

$$k^2 - \frac{n^2}{R_i^2} = 0. \quad (13)$$

which gives the required result.

2. Chemotaxis

(a) The three Laplacians represent the diffusive motion of all the populations.

The term

$$-\alpha \nabla \left[\frac{n}{(1+c)^2} \nabla c \right], \quad (14)$$

represents the attraction of the bacteria n to each other due to the release of the chemoattractant c .

The term

$$\rho n \left(\delta \frac{s^2}{1+s^2} - n \right) \quad (15)$$

represents logistic type growth $n(K-n)$ of the bacteria population. The equilibrium size K is determined by the nutrient population s and this term is bound between 0 and δ .

The term

$$\beta s \frac{n^2}{\mu + n^2} \quad (16)$$

represents the increase in chemoattractant due to the stimulant s , and β is the rate of growth. It also depends on the bacterial density n indicating as the cells produce the chemoattractant as a result of consuming the nutrient. Its dependence on n indicates that it saturates as a function of n .

The term

$$-nc \quad (17)$$

represents the consumption of the chemoattractant by the cells.
Finally the term

$$-\kappa n \frac{s^2}{1 + s^2}, \quad (18)$$

represents the consumption of the nutrient s by the bacteria, the s dependence indicated the consumption per density element saturates (the bacterial cells can only consume so much nutrient in a given time frame).

- (b) The homogeneous equilibria are $(n = 0, c = c_0, s = s_0)$.
- (c) If $n_0 \neq 0$ (homogeneous) then s_0 must be positive, the the second term of the s equation must be non-zero and s would not be in equilibrium.

3. Chemotaxis and slime mould (Exam section B, 2017)

- (a) The three Laplacians represent the diffusive motion of all the populations.

The term

$$-\alpha \nabla \left[\frac{n}{(1 + c)^2} \nabla c \right] \quad (19)$$

represents the attraction of the bacteria n to each other due to the release of the chemoattractant c .

The term

$$\rho n \left(\delta \frac{s^2}{1 + s^2} - n \right) \quad (20)$$

represents logistic type growth $n(K - n)$ of the bacteria population. The equilibrium size K is determined by the nutrient population s and this term is bound between 0 and δ .

The term

$$\beta s \frac{n^2}{\mu + n^2} \quad (21)$$

represents the increase in chemoattractant due to the stimulant s , β is the rate if growth. It also depends on the bacterial density n indicating as the bacteria produce the chemoattractant as a result of consuming the nutrient. Its dependence on n indicates that it saturates as a function of n .

The term

$$-nc \quad (22)$$

represents the consumption of the chemoattractant by the cells.

Finally the term

$$-\kappa n \frac{s^2}{1 + s^2} \quad (23)$$

represents the consumption of the nutrient s by the bacteria, the s dependence indicated the consumption per density element saturates (the bacterial cells can only consume so much nutrient in a given time frame).

- (b) $D > 1$, this is required so that the chemoattractant can attract bacterial cells to it.

(c) The functions are

$$F(n, c) = \rho n \left(\delta \frac{s^2}{1 + s^2} - n \right), \quad (24)$$

$$G(n, c) = \beta s \frac{n^2}{\mu + n^2} - nc, \quad (25)$$

$$\chi(n, c) = \frac{n}{(1 + c)^2}. \quad (26)$$

(d) Writing

$$S_0 = \delta \frac{s_0^2}{1 + s_0^2}, \quad (27)$$

the equilibria are

$$F(n_0, c_0) = 0, \quad G(n_0, c_0) = 0. \quad (28)$$

The first equation $F(n_0, c_0) = 0$ is solved if $n = 0, S_0$. In the first case, $G(0, c_0) = 0$ places no constraints on c_0 . In the second case,

$$c_0 = \beta s_0 \frac{S_0}{\mu + S_0^2}. \quad (29)$$

(e) We have

$$F_n = \rho(S_0 - 2n_0), \quad (30)$$

$$F_c = 0, \quad (31)$$

$$G_n = \beta s_0 \left[\frac{2n_0}{\mu + n_0^2} - \frac{2n_0^3}{(\mu + n_0^2)^2} \right] - c_0, \quad (32)$$

$$G_c = -n_0. \quad (33)$$

So for the case $n_0 = 0$ the first condition is

$$F_n + G_c < 0 \implies \rho S_0 < 0. \quad (34)$$

But this is cannot be true as $\rho > 0$ and $S_0 > 0$.

For the second equilibrium $n_0 = S_0$, the first condition becomes

$$F_n + G_c < 0 \implies -S_0(\rho + 1) < 0, \quad (35)$$

which is satisfied for all permissible ρ and S_0 . The second condition in this case implies

$$F_n G_c > 0 \implies \rho S_0^2 > 0 \quad (36)$$

which is also satisfied for all permissible ρ and S_0 . For the third condition we will also need G_n simplified. Using $n_0 = S_0$ we have

$$G_n = \frac{2\beta s_0 S_0}{\mu + S_0^2} \left[1 - \frac{S_0^2}{\mu + S_0^2} \right] - \beta s_0 \frac{S_0}{\mu + S_0^2} = \frac{\beta s_0 S_0}{\mu + S_0^2} \left[1 - \frac{2S_0^2}{\mu + S_0^2} \right]. \quad (37)$$

The third condition then reads

$$G_c + D F_n - \alpha \chi_0 G_n > 0 \quad (38)$$

$$\implies -S_0 - D\rho S_0 - \alpha \chi_0 \frac{\beta s_0 S_0}{\mu + S_0^2} \left[1 - \frac{2S_0^2}{\mu + S_0^2} \right] > 0 \quad (39)$$

$$\implies -S_0 \left\{ 1 + D\rho + \alpha \chi_0 \frac{\beta s_0}{\mu + S_0^2} \left[1 - \frac{2S_0^2}{\mu + S_0^2} \right] \right\} > 0. \quad (40)$$

So we require

$$\alpha\chi_0\frac{\beta s_0}{\mu + S_0^2}\left[1 - \frac{2S_0^2}{\mu + S_0^2}\right] < -(1 + D\rho). \quad (41)$$

As S_0 is bound monotonically between 0 and δ , the difference

$$1 - \frac{2S_0^2}{\mu + S_0^2}, \quad (42)$$

cannot be negative if

$$\mu > \delta^2. \quad (43)$$

As χ_0, α, β and s_0 are positive this means the condition cannot be satisfied.

4. Non-Euclidean domain

- (a) The θ boundary conditions are necessarily periodic,

$$u(r, 0, t) = u(r, 2\pi, t), \text{ and } v(r, 0, t) = v(r, 2\pi, t). \quad (44)$$

The r boundary condition is

$$\frac{\partial u}{\partial r}(R_o, \theta, t) = 0, \quad (45)$$

and similar for v .

- (b) This is just bookwork from your notes. In an exam you could assume the Fourier series solution just by citing the periodicity of the domain in the θ coordinates.
- (c) I will leave the argument to you as to why, but you should be choosing the $n = 0$ modes only so the sin parts are automatically zero. We note that the function Y_0 will blow up as $r \rightarrow 0$ so the constant B_{01} must be zero (why don't I care about $B_{02} \dots$?). Thus we have

$$\psi_u(r, \theta) = A_{01}J_0(kr). \quad (46)$$

Now, to get an equation for k we must apply the radial boundary condition (we have of course already enforced the θ boundary conditions). From the form of J_0 we see $dJ_0(kr)/dr = -kJ_1(kr)$. So applying $du/dr = 0$ at $r = R_o$ yields the required equation.

- (d) This relies on your underlying understanding of the Turing mechanism. Remember that satisfying the Turing conditions implies there is some finite domain $k \in [k_{\min}, k_{\max}]$ for which patterns will grow in time ($\lambda > 0$). Increasing R_0 means more zeros of J_1 in the same k domain $k \in [k_{\min}, k_{\max}]$, thus more likely patterns which are assumed to form due to random changes about the homogeneous equilibrium. So the $R_0 = 100$ case should have more patterns all else being equal.

5. Chemotaxis (Exam section B type)

- (a) If c is not spatially varying then $\partial c/\partial x = 0$ so n can be any constant. The equation for c then reduces to

$$\frac{dc}{dt} = \gamma c \frac{n_0^2}{\mu + n_0^2}, \quad (47)$$

whose solution is

$$c = Ae^{rt}, \quad r = \gamma \frac{n_0^2}{\mu + n_0^2}. \quad (48)$$

The initial condition c_0 sets A and gives the required answer.

(b) To linear order,

$$\begin{aligned} \frac{\partial n}{\partial t} &= \varepsilon \frac{\partial f}{\partial t} \sum_k e^{ikx}, & \frac{\partial c}{\partial t} &= rc_0 e^{rt} + \varepsilon \frac{\partial g}{\partial t} \sum_k e^{ikx}, \\ \nabla^2 n &= -\varepsilon k^2 f(t) \sum_k e^{ikx}, & \nabla^2 c &= -\varepsilon k^2 g(t) \sum_k e^{ikx}, \\ \frac{\partial c}{\partial x} &= -\varepsilon i k g(t) \sum_k e^{ikx}, & n \frac{\partial c}{\partial x} &\approx \varepsilon n_0 i k g(t) \sum_k e^{ikx} + \mathcal{O}(\varepsilon^2), \\ & & \frac{\partial}{\partial x} \left[n \frac{\partial c}{\partial x} \right] &\approx -\varepsilon n_0 k^2 g(t) \sum_k e^{ikx} + \mathcal{O}(\varepsilon^2). \end{aligned} \quad (49)$$

The last term needed is a little tricky. If we just denote temporarily n_0 and c_0^* as the zeroth order terms and n_1, c_1 the corrections

$$c \frac{n^2}{\mu + n^2} \approx c_0^* \frac{n_0^2}{\mu + n_0^2} + \varepsilon \left[n_1 c_0^* \left(\frac{2n_0}{\mu + n_0^2} - \frac{2n_0^3}{(\mu + n_0^2)^2} \right) + c_1 \frac{n_0^2}{\mu + n_0^2} \right]. \quad (50)$$

then using our definition of r the $\mathcal{O}(\varepsilon)$ correction is

$$\frac{2n_1 n_0 c_0^*}{\mu + n_0^2} \left(1 - \frac{r}{\gamma} \right) + c_1 \frac{r}{\gamma} \quad (51)$$

and substituting back in c_0^*, n_1, c_1 the order ε term is

$$\frac{2n_0 c_0}{\mu + n_0^2} \left(1 - \frac{r}{\gamma} \right) e^{rt} f(t) \sum_k e^{ikx} + \frac{r}{\gamma} g(t) \sum_k e^{ikx}. \quad (52)$$

Collecting all $\mathcal{O}(\varepsilon)$ terms we find for each k that

$$\frac{\partial f}{\partial t} = -Dk^2 f + \alpha n_0 k^2 g, \quad (53)$$

$$\frac{\partial g}{\partial t} = -(k^2 - r)g + \frac{2n_0 c_0}{\mu + n_0^2} (\gamma - r) e^{rt} f. \quad (54)$$

(c) Taking the derivatives of the first equation we get

$$\frac{\partial^2 f}{\partial t^2} = -Dk^2 \frac{\partial f}{\partial t} + \alpha n_0 k^2 \frac{\partial g}{\partial t}. \quad (55)$$

We then substitute in the second equation to find

$$\frac{\partial^2 f}{\partial t^2} = -Dk^2 \frac{\partial f}{\partial t} + \alpha n_0 k^2 \{ -(k^2 - r)g + \beta (\gamma - r) e^{rt} f \}, \quad (56)$$

$$\implies \frac{\partial^2 f}{\partial t^2} = -Dk^2 \frac{\partial f}{\partial t} + \alpha n_0 \beta (\gamma - r) e^{rt} k^2 f - \alpha n_0 k^2 (k^2 - r) g, \quad (57)$$

where $\beta = \frac{2n_0 c_0}{\mu + n_0^2}$. We now use the first equation again to obtain an expression for g ,

$$\alpha n_0 k^2 g = \frac{\partial f}{\partial t} + D k^2 f, \quad (58)$$

so that

$$\frac{\partial^2 f}{\partial t^2} = -[D k^2 + (k^2 - r)] \frac{\partial f}{\partial t} + [\alpha n_0 \beta (\gamma - r) e^{rt} - (k^2 - r) D] k^2 f. \quad (59)$$

- (d) The coefficient of the $f(t)$ part of the equation contains an exponential term. Since we would expect the constant coefficient case to show exponential growth or decay the assumption of differing rates of change is not valid.
- (e) In this case $b(k) = 0$. We have an equation with constant coefficients. We seek a solution in the form $e^{\lambda t}$ for which

$$\lambda^2 + a\lambda + c = 0, \quad (60)$$

so

$$\lambda = \frac{1}{2} \left[-a \pm \sqrt{a^2 - 4c} \right]. \quad (61)$$

If

$$k^2 - r > 0, \quad (62)$$

then c and a will be positive, and both eigenvalues will have a negative real part. The no flux boundary conditions imply $k = n\pi/L$ so we need the lowest mode $n = 1$ to be stable, hence

$$\frac{\pi^2}{L^2} - r > 0. \quad (63)$$

6. Dynamic Turing analysis

No solution provided.